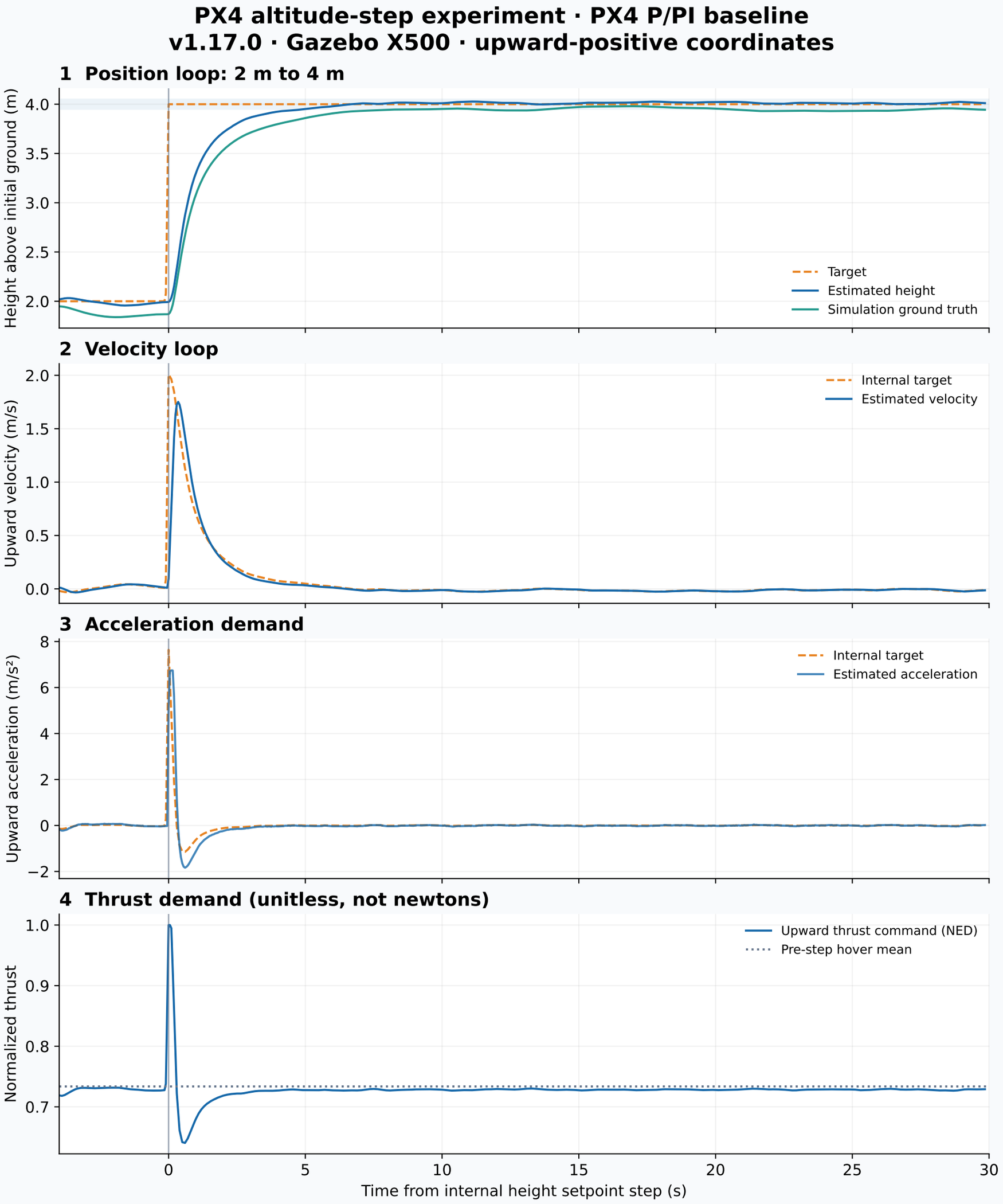
PX4 altitude-step experiment · PX4 P/PI baseline
v1.17.0 · Gazebo X500 · upward-positive coordinates
1 Position loop: 2 m to 4 m
Height above initial ground (m)
Target
Estimated height
Simulation ground truth
2 Velocity loop
Upward velocity (m/s)
Internal target
Estimated velocity
3 Acceleration demand
Upward acceleration (m/s²)
Internal target
Estimated acceleration
4 Thrust demand (unitless, not newtons)
Normalized thrust
Upward thrust command (NED)
Pre-step hover mean
Time from internal height setpoint step (s)